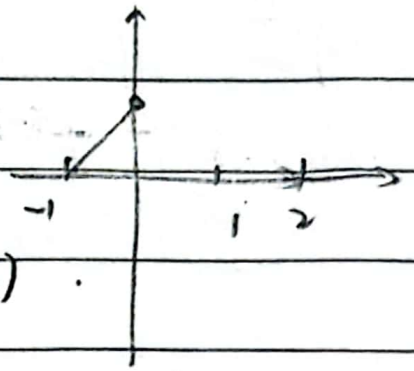
思考题四 6 7 10

习题四 4(1)(4)15) 5(1) 10 12 13(2)(6)

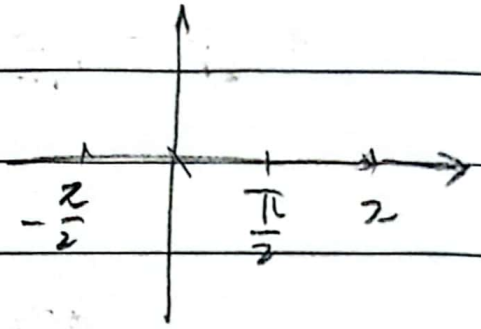
6、 $R=\sqrt{2}$



7、 $\cos z=0 \Rightarrow e^{iz}=e^{-iz}=e^{i(\pi-z)}$

$z=\pi-z+2k\pi \Rightarrow z=\frac{\pi}{2}+k\pi$

$R=\frac{\pi}{2}$



10 $f(z)=\frac{1}{z(z^2-1)}$

$z=0$, $R_1<|z|<R_2$, $0\le R_1<R_2\le 1$, $\sum_{n=-\infty}^{+\infty} C_n z^n$

或 $R_1\ge 1$



$z=1$, $R_1<|z-1|<R_2$, $0\le R_1<R_2\le 1$, $\sum_{-\infty}^{+\infty} C_n(z-1)^n$

或 $1\le R_1<R_2\le 2$, 或 $R_1\ge 2$

$z=-1$, $R_1<|z+1|<R_2$, $0\le R_1<R_2\le 1$, $\sum_{-\infty}^{+\infty} C_n(z+1)^n$

或 $1\le R_1<R_2\le 2$ 或 $R_1\ge 2$

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$\sum x^n = \frac{1}{1-x}$ $\sum (-1)^n x^n = \frac{1}{1+x}$

$\sum n x^{n-1} = (\frac{1}{1-x})'$

4 (1) $\frac{1}{z+1} = \frac{1}{2+z-1}$

$= \frac{1}{2} \frac{1}{1+\frac{z-1}{2}} = \frac{1}{2} \sum (-1)^n \frac{(z-1)^n}{2^n}$

$= \sum (-1)^n \frac{(z-1)^{n+1}}{2^{n+1}}$

$|\frac{z-1}{2}| < 1$

(4) $\frac{1}{(z-1)^2} = \frac{1}{[1-(z-1)]^2} = [\frac{1}{1-(z-1)}]' = [\sum_{n=0}^{\infty} (z-1)^n]'$

$= \sum_{n=0}^{\infty} n(z-1)^{n-1}$ $(|z-1|<1)$

$|z-1|<1$

(5) $\frac{1}{z^2-2z+10} = \frac{1}{(z-1)^2+9} = \frac{1}{9} \frac{1}{(\frac{z-1}{3})^2+1} = \frac{1}{9} \sum (-1)^n \frac{(z-1)^{2n}}{3^{2n}}$

$|(\frac{z-1}{3})^2| < 1 \Rightarrow |z-1| < 3$

5(1) $\frac{1}{(1+x)^2} = \sum_{n=0}^{\infty} n(-1)^n x^{n-1}$

故 $\frac{1}{(1+z^2)^2} = \sum_{n=0}^{\infty} n(-1)^n z^{2n-2}$ $|z^2|<1 \Rightarrow |z|<1$

10. $f(z) = e^{z^2} = 1 + z^2 + \frac{z^4}{2} + \cdots + \frac{z^{2n}}{n!} + \cdots$

故 $f^{(n)}(0) = \frac{(2n)!}{n!}$

12. (1) $f^{(n)}(z_0) = \frac{n!}{2\pi i}\int \frac{f(z)}{(z-z_0)^{n+1}} dz$

$f(\zeta) = e^{z\zeta}$, $\zeta_0 = 0$, 则 $z^n = \frac{n!}{2\pi i}\int \frac{e^{z\zeta}}{\zeta^{n+1}} d\zeta$

$\Rightarrow \left(\frac{z^n}{n!}\right)^2 = \frac{1}{2\pi i}\oint_{|\zeta|=1} \frac{z^n e^{z\zeta}}{n!\zeta^n}\frac{d\zeta}{\zeta}$

(2) $\sum_{n=0}^{\infty}\left(\frac{z^n}{n!}\right)^2 = \frac{1}{2\pi i}\oint_{|\zeta|=1} \frac{\sum_{n=0}^{\infty}\frac{z^n}{n!\zeta^n}e^{z\zeta}}{\zeta} d\zeta = \frac{1}{2\pi i}\oint_{|\zeta|=1} \frac{e^{\frac{z}{\zeta}}e^{z\zeta}}{\zeta} d\zeta$

令 $\zeta = e^{i\theta}$, 则 $\sum_{n=0}^{\infty}\left(\frac{z^n}{n!}\right)^2 = \frac{1}{2\pi}\int_0^{2\pi} e^{2z\cos\theta} d\theta$

$= \cos\theta + i\sin\theta$

13 (a) $|z-1|>1 \Rightarrow 0<\left|\frac{1}{z-1}\right|<1$



故 $\frac{1}{1+\frac{1}{z-1}} = \sum_{n=0}^{\infty}(-1)^n \frac{1}{(z-1)^n}$

$\Rightarrow \frac{1}{z} = \sum_{n=0}^{\infty}(-1)^n (z-1)^{-n-1}$

$-\frac{1}{z^2} = \sum_{n=0}^{\infty}(-1)^n(-n-1)(z-1)^{-n-2}$

$\frac{1}{z^2} = \sum_{n=0}^{\infty}(-1)^n(n+1)(z-1)^{-n-2}$

故 $\frac{z-1}{z^2} = \sum_{n=0}^{\infty}(-1)^n(n+1)(z-1)^{-n-1} \quad (|z-1|>1)$

(b) $\frac{1}{z} = \frac{1}{z+2-2} = -\frac{1}{2}\frac{1}{1-\frac{z+2}{2}} = -\frac{1}{2}\sum_{n=0}^{\infty}\left(\frac{z+2}{2}\right)^n = \sum_{n=0}^{\infty} -\frac{(z+2)^n}{2^{n+1}}$

$\frac{1}{z(z+2)^3} = \sum_{n=0}^{\infty} -\frac{(z+2)^{n-3}}{2^{n+1}} \quad (0<|z+2|<2)$

思考题五 2,7

习题五 1(1)(6)(7) 2(4)(6)

思考题五

T2 不可以，$z=0$ 不为 $f(z)=\frac{1}{(z-1)(z-2)}$ 的孤立奇点，

T7 不一定，

$f(z)=\sin\frac{1}{z^2}=\frac{1}{z^2}-\frac{1}{3!z^6}+\cdots+\frac{(-1)^n}{(2n+1)!}\frac{1}{z^{4n+2}}+\cdots$

$\operatorname{Res}[f(z),0]=C_{-1}=0$

而 $z_0=0$ 为 $f(z)$ 的本性奇点.

习题五

T1 (1) 对 $f(z)=z(z^2+1)^2$

$f'(z)=(z^2+1)^2+4z^2(z^2+1)=(z^2+1)(5z^2+1)$

$f''(z)=2z(5z^2+1)+(z^2+1)10z=20z^3+12z$

对 $z=0$，$f(z)=0$，$f'(z)=1\neq 0$，$z=0$ 为原式一级极点

对 $z=\pm i$，$f(z)=f'(z)=0$，$f''(z)=\mp 8i\neq 0$，$z=\pm i$ 为原式二级极点

(6) $f(z)=e^{\frac{1}{z-1}}$ $\lim\limits_{z\to1^+}f(z)=+\infty$ $\lim\limits_{z\to1^-}f(z)=0$

$\lim\limits_{z\to1}f(z)$ 不存在，故 $z=1$ 为 $e^{\frac{1}{z-1}}$ 本性奇点

(7) $f(z)=e^{z^2}$ $f(1)=e\neq0$

$g(z)=(z-1)^2$ $z=1$ 为 $g(z)$ 二级零点

故 $z=1$ 为 $\frac{e^{z^2}}{(z-1)^2}$ 二级极点

2. (4) $z=0$

$$\frac{1+e^z}{z^2}+\frac{1}{z}=\frac{2+z+\frac{z^2}{2!}+\frac{z^3}{3!}+\cdots}{z^2}+\frac{1}{z}=\frac{2}{z^2}+\frac{2}{z}+\frac{1}{2}+\frac{z^3}{3!}+\cdots$$

$$\operatorname{Res}\left(\frac{1+e^z}{z^2}+\frac{1}{z};0\right)=C_{-1}=2$$

(6) $f(z)=z\sin z$ $f'(z)=\sin z+z\cos z$ $f''(z)=2\cos z-z\sin z$

故 $z=0$ 为 $\frac{1}{z\sin z}$ 二级极点，$z=k\pi$，$k\in\mathbb{Z}$ 且 $k\neq0$ 为 $\frac{1}{z\sin z}$ 单极点，

对 $z=k\pi$，$\operatorname{Res}\left[\frac{1}{z\sin z};k\pi\right]=\frac{1}{\sin z+z\cos z}\Big|_{z=k\pi}=(-1)^k\frac{1}{k\pi}$

对 $z=0$，$\operatorname{Res}\left[\frac{1}{z\sin z};k\pi\right]=\lim\limits_{z\to0}\left(\frac{z}{\sin z}\right)'=\lim\limits_{z\to0}\frac{\sin z-z\cos z}{\sin^2 z}=0$

偶函数，不含奇数次幂